Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20160118-004-DS-6-RA-103	Issued	2016-02-12

UT Thickness Data Sheet - 6-RA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the rich amine header 6-RA-103 located within the ATU-100. The scope of work is limited to the thickness survey of this specific line to support the asset integrity program.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis

All thickness measurements were performed in accordance with QSCP-NDT-001 Rev 2. This procedure incorporates specific requirements for the CUI program, including the management of insulation removal and reinstatement and the use of inspection ports.

The measurement techniques and equipment calibration are based on the requirements of ASTM E797. The data collected for this line is integrated into the broader survey findings detailed in LNDE-RPT-20160118-004-R0.

Line Specifications

The subject line, 6-RA-103, is constructed from A106 Gr.B with a nominal outer diameter of 168.3 mm and a nominal wall thickness of 10.97 mm. The design limits for this service are 862 kPa(g) and 82 ?C.

The minimum allowable thickness (t-min) for this line is 4.57 mm. Any reading recorded below this retirement thickness shall be raised as an NCR.

Inspection Interval and Classification

As a rich amine header within the ATU-100, 6-RA-103 is subject to a 3-year thickness measurement interval. This is consistent with the requirements for rich amine and regenerator overhead circuits.

The line is categorized under Class 2 for amine service, which typically allows a 10-year thickness and 5-year external visual interval, though the specific 3-year rich amine requirement takes precedence for this survey.

Thickness readings, measured 2016-02-12

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2011-02-18	6.39	6.17	0.048	0.044	>50	2021-02-12	
6-SG-101-02	7.11	3.56	2011-02-18	6.14	5.76	0.065	0.076	28.8	2021-02-12	
6-SG-101-03	7.11	3.56	2011-02-18	6.17	5.87	0.061	0.060	38.1	2021-02-12	
6-SG-101-04	7.11	3.56	2011-02-18	6.05	5.58	0.076	0.094	21.4	2021-02-12	
6-SG-101-05	7.11	3.56	2011-02-18	5.87	5.38	0.088	0.098	18.5	2021-02-12	
6-SG-101-06	7.11	3.56	2011-02-18	6.27	5.88	0.060	0.078	29.6	2021-02-12	
6-SG-101-08	7.11	3.56	2011-02-18	6.33	6.00	0.066	0.066	36.8	2021-02-12	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2011-02-18	5.75	5.42	0.086	0.066	21.6	2021-02-12	
6-SG-102-02	7.11	3.56	2011-02-18	5.80	5.43	0.084	0.074	22.2	2021-02-12	
6-SG-102-03	7.11	3.56	2011-02-18	5.80	5.22	0.091	0.116	14.3	2021-02-12	
6-SG-102-04	7.11	3.56	2011-02-18	5.73	5.18	0.091	0.110	14.7	2021-02-12	
6-SG-102-05	7.11	3.56	2011-02-18	6.46	6.19	0.047	0.054	48.5	2021-02-12	
6-SG-102-06	7.11	3.56	2011-02-18	6.41	6.06	0.070	0.070	35.6	2021-02-12	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2006-02-24	7.89	7.76	0.022	0.013	>50	2026-02-12	
8-TG-101-02	8.18	4.06	2006-02-24	7.75	7.46	0.037	0.029	>50	2026-02-12	
8-TG-101-03	8.18	4.06	2006-02-24	7.79	7.54	0.035	0.025	>50	2026-02-12	
8-TG-101-04	8.18	4.06	2006-02-24	7.76	7.51	0.030	0.025	>50	2026-02-12	
8-TG-101-05	8.18	4.06	2006-02-24	7.95	7.73	0.025	0.022	>50	2026-02-12	
8-TG-101-07	8.18	4.06	—	—	7.39	—	—	—	2026-02-12	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2006-02-24	5.51	5.02	0.049	0.049	37.4	2026-02-12	
4-LA-101-02	6.02	3.18	2006-02-24	5.76	5.45	0.028	0.031	>50	2026-02-12	
4-LA-101-03	6.02	3.18	2006-02-24	5.76	5.45	0.026	0.031	>50	2026-02-12	
4-LA-101-04	6.02	3.18	2006-02-24	5.78	5.48	0.030	0.030	>50	2026-02-12	
4-LA-101-05	6.02	3.18	2006-02-24	5.81	5.54	0.024	0.027	>50	2026-02-12	
4-LA-101-06	6.02	3.18	2006-02-24	5.64	5.30	0.038	0.034	>50	2026-02-12	
4-LA-101-07	6.02	3.18	2006-02-24	5.57	4.87	0.056	0.070	24.1	2026-02-12	
4-LA-101-09	6.02	3.18	—	—	5.00	—	—	—	2026-02-12	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2006-02-24	5.63	5.25	0.039	0.038	>50	2026-02-12	
4-LA-102-02	6.02	3.18	2006-02-24	5.67	5.36	0.036	0.031	>50	2026-02-12	
4-LA-102-03	6.02	3.18	2006-02-24	5.55	5.28	0.040	0.027	>50	2026-02-12	
4-LA-102-04	6.02	3.18	2006-02-24	5.34	4.90	0.055	0.044	31.5	2026-02-12	
4-LA-102-05	6.02	3.18	2006-02-24	5.62	5.34	0.033	0.028	>50	2026-02-12	
4-LA-102-06	6.02	3.18	2006-02-24	5.83	5.60	0.020	0.023	>50	2026-02-12	
4-LA-102-07	6.02	3.18	2006-02-24	5.63	5.40	0.030	0.023	>50	2026-02-12	
4-LA-102-08	6.02	3.18	—	—	5.70	—	—	—	2026-02-12	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2006-02-24	6.75	6.44	0.029	0.031	>50	2026-02-12	
6-LA-103-02	7.11	3.56	2006-02-24	6.87	6.53	0.033	0.034	>50	2026-02-12	
6-LA-103-03	7.11	3.56	2006-02-24	6.65	6.28	0.040	0.037	>50	2026-02-12	
6-LA-103-04	7.11	3.56	2006-02-24	6.52	6.15	0.051	0.037	>50	2026-02-12	
6-LA-103-05	7.11	3.56	2006-02-24	6.49	5.98	0.061	0.051	39.9	2026-02-12	
6-LA-103-06	7.11	3.56	2006-02-24	6.42	5.82	0.067	0.060	33.6	2026-02-12	
6-LA-103-07	7.11	3.56	2006-02-24	6.76	6.12	0.053	0.064	39.9	2026-02-12	
6-LA-103-08	7.11	3.56	2006-02-24	6.79	6.49	0.028	0.030	>50	2026-02-12	
6-LA-103-10	7.11	3.56	—	—	6.30	—	—	—	2026-02-12	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2014-09-19	8.07	7.83	0.035	0.098	40.9	2019-02-12	
4-RA-101-02	8.56	3.81	2014-09-19	8.02	7.94	0.030	0.052	>50	2019-02-12	
4-RA-101-03	8.56	3.81	2014-09-19	7.88	7.65	0.048	0.066	>50	2019-02-12	
4-RA-101-04	8.56	3.81	2014-09-19	8.18	8.00	0.032	0.042	>50	2019-02-12	
4-RA-101-05	8.56	3.81	2014-09-19	7.98	7.82	0.039	0.068	>50	2019-02-12	
4-RA-101-06	8.56	3.81	2014-09-19	7.91	7.87	0.029	0.029	>50	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-07	8.56	3.81	2014-09-19	8.22	8.14	0.023	0.032	>50	2019-02-12	
4-RA-101-08	8.56	3.81	2014-09-19	7.71	7.31	0.069	0.142	24.6	2019-02-12	
4-RA-101-09	8.56	3.81	2014-09-19	8.27	8.13	0.022	0.038	>50	2019-02-12	
4-RA-101-10	8.56	3.81	2014-09-19	8.21	8.02	0.033	0.052	>50	2019-02-12	
4-RA-101-11	8.56	3.81	2014-09-19	7.88	7.60	0.054	0.088	42.9	2019-02-12	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2014-09-19	8.25	8.02	0.026	0.062	>50	2019-02-12	
4-RA-102-02	8.56	3.81	2014-09-19	7.89	7.64	0.047	0.070	>50	2019-02-12	
4-RA-102-03	8.56	3.81	2014-09-19	7.95	7.55	0.051	0.098	38.0	2019-02-12	
4-RA-102-04	8.56	3.81	2014-09-19	7.93	7.72	0.041	0.048	>50	2019-02-12	
4-RA-102-05	8.56	3.81	2014-09-19	7.84	7.67	0.044	0.080	48.1	2019-02-12	
4-RA-102-06	8.56	3.81	2014-09-19	7.95	7.73	0.039	0.056	>50	2019-02-12	
4-RA-102-07	8.56	3.81	2014-09-19	7.98	7.78	0.040	0.064	>50	2019-02-12	
4-RA-102-08	8.56	3.81	2014-09-19	7.58	7.35	0.057	0.106	33.3	2019-02-12	
4-RA-102-09	8.56	3.81	2014-09-19	7.84	7.54	0.051	0.088	42.2	2019-02-12	
4-RA-102-11	8.56	3.81	2014-09-19	8.12	7.96	0.046	0.046	>50	2019-02-12	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2014-09-19	10.62	10.29	0.036	0.036	>50	2019-02-12	
6-RA-103-02	10.97	4.57	2014-09-19	10.06	9.69	0.070	0.108	47.2	2019-02-12	
6-RA-103-03	10.97	4.57	2014-09-19	10.39	10.06	0.047	0.072	>50	2019-02-12	
6-RA-103-04	10.97	4.57	2014-09-19	9.79	9.31	0.084	0.132	35.8	2019-02-12	
6-RA-103-05	10.97	4.57	2014-09-19	9.81	9.48	0.074	0.124	39.5	2019-02-12	
6-RA-103-06	10.97	4.57	2014-09-19	10.43	10.32	0.035	0.048	>50	2019-02-12	
6-RA-103-07	10.97	4.57	2014-09-19	10.37	10.20	0.040	0.070	>50	2019-02-12	
6-RA-103-08	10.97	4.57	2014-09-19	10.49	10.33	0.032	0.022	>50	2019-02-12	
6-RA-103-09	10.97	4.57	2014-09-19	10.39	10.33	0.034	0.046	>50	2019-02-12	
6-RA-103-10	10.97	4.57	2014-09-19	9.93	9.49	0.077	0.112	43.8	2019-02-12	
6-RA-103-11	10.97	4.57	2014-09-19	10.10	9.77	0.236	0.236	22.0	2019-02-12	
6-RA-103-13	10.97	4.57	2014-09-19	10.38	10.15	0.060	0.060	>50	2019-02-12	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2013-09-09	6.49	6.34	0.063	0.081	34.2	2019-02-12	
3-RV-101-02	7.62	3.56	2013-09-09	6.44	6.31	0.063	0.052	43.5	2019-02-12	
3-RV-101-03	7.62	3.56	2013-09-09	5.80	5.63	0.095	0.098	21.2	2019-02-12	
3-RV-101-04	7.62	3.56	2013-09-09	6.13	5.88	0.086	0.102	22.8	2019-02-12	
3-RV-101-05	7.62	3.56	2013-09-09	6.48	6.36	0.066	0.050	42.6	2019-02-12	
3-RV-101-06	7.62	3.56	2013-09-09	6.11	5.96	0.082	0.087	27.7	2019-02-12	
3-RV-101-07	7.62	3.56	2013-09-09	5.73	5.54	0.122	0.122	16.2	2019-02-12	
3-RV-101-08	7.62	3.56	2013-09-09	6.36	6.31	0.074	0.074	37.2	2019-02-12	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2013-09-09	4.49	4.35	0.059	0.037	26.4	2019-02-12	
2-RV-102-02	5.54	2.79	2013-09-09	4.29	4.01	0.079	0.109	11.2	2019-02-12	
2-RV-102-03	5.54	2.79	2013-09-09	4.68	4.64	0.042	0.037	43.9	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2013-09-09	4.81	4.73	0.043	0.020	45.5	2019-02-12	
2-RV-102-05	5.54	2.79	2013-09-09	4.96	4.97	0.013	0.013	>50	2019-02-12	
2-RV-102-06	5.54	2.79	2013-09-09	4.55	4.27	0.102	0.102	14.6	2019-02-12	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-12-09	5.37	5.31	0.009	0.017	>50	2026-02-12	
3-RF-101-02	5.49	2.41	2013-12-09	5.40	5.34	0.006	0.003	>50	2026-02-12	
3-RF-101-03	5.49	2.41	2013-12-09	5.37	5.33	0.006	0.012	>50	2026-02-12	
3-RF-101-04	5.49	2.41	2013-12-09	5.41	5.35	0.006	-0.002	>50	2026-02-12	
3-RF-101-05	5.49	2.41	2013-12-09	5.32	5.33	0.006	0.007	>50	2026-02-12	
3-RF-101-06	5.49	2.41	2013-12-09	5.17	5.23	0.015	0.017	>50	2026-02-12	
3-RF-101-07	5.49	2.41	2013-12-09	5.48	5.37	0.003	-0.002	>50	2026-02-12	
3-RF-101-09	5.49	2.41	2013-12-09	5.26	5.19	-0.029	-0.029	>50	2026-02-12	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-12-09	3.60	3.76	0.008	0.009	>50	2026-02-12	
2-RF-102-02	3.91	2.16	2013-12-09	3.70	3.64	0.012	0.024	>50	2026-02-12	
2-RF-102-03	3.91	2.16	2013-12-09	3.79	3.75	0.009	0.005	>50	2026-02-12	
2-RF-102-04	3.91	2.16	2013-12-09	3.82	3.38	0.025	0.043	28.3	2026-02-12	
2-RF-102-05	3.91	2.16	2013-12-09	3.67	3.49	0.082	0.082	16.1	2024-03-06	
2-RF-102-06	3.91	2.16	2013-12-09	3.67	3.47	0.099	0.099	13.2	2022-09-26	
2-RF-102-07	3.91	2.16	2013-12-09	3.72	3.57	0.020	0.028	>50	2026-02-12	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2006-02-24	5.75	5.31	0.038	0.044	48.2	2026-02-12	
4-LA-104-02	6.02	3.18	2006-02-24	5.56	5.00	0.052	0.056	32.4	2026-02-12	
4-LA-104-03	6.02	3.18	2006-02-24	5.74	5.42	0.027	0.032	>50	2026-02-12	
4-LA-104-04	6.02	3.18	2006-02-24	5.86	5.63	0.022	0.023	>50	2026-02-12	
4-LA-104-05	6.02	3.18	2006-02-24	5.51	4.88	0.058	0.063	26.9	2026-02-12	
4-LA-104-06	6.02	3.18	2006-02-24	5.83	5.55	0.028	0.028	>50	2026-02-12	
4-LA-104-07	6.02	3.18	2006-02-24	5.62	5.28	0.037	0.034	>50	2026-02-12	
4-LA-104-08	6.02	3.18	2006-02-24	5.50	5.11	0.042	0.039	46.3	2026-02-12	
4-LA-104-10	6.02	3.18	—	—	4.84	—	—	—	2026-02-12	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2014-09-19	7.84	7.71	0.040	0.048	>50	2019-02-12	
4-RA-104-02	8.56	3.81	2014-09-19	8.09	7.85	0.035	0.058	>50	2019-02-12	
4-RA-104-03	8.56	3.81	2014-09-19	8.46	8.28	0.057	0.056	>50	2019-02-12	
4-RA-104-04	8.56	3.81	2014-09-19	8.49	8.18	0.063	0.084	>50	2019-02-12	
4-RA-104-05	8.56	3.81	2014-09-19	8.37	8.16	0.050	0.082	>50	2019-02-12	
4-RA-104-06	8.56	3.81	2014-09-19	7.91	7.61	0.049	0.074	>50	2019-02-12	
4-RA-104-07	8.56	3.81	2014-09-19	7.61	7.32	0.064	0.092	38.0	2019-02-12	
4-RA-104-08	8.56	3.81	2014-09-19	8.00	7.81	0.074	0.074	>50	2019-02-12	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2006-02-24	3.64	3.61	0.016	0.003	>50	2026-02-12	
2-ST-101-02	3.91	2.29	2006-02-24	3.67	3.55	0.018	0.012	>50	2026-02-12	
2-ST-101-03	3.91	2.29	2006-02-24	3.82	3.55	0.020	0.027	46.5	2026-02-12	
2-ST-101-04	3.91	2.29	2006-02-24	3.77	3.68	0.007	0.009	>50	2026-02-12	
2-ST-101-05	3.91	2.29	2006-02-24	3.72	3.50	0.021	0.022	>50	2026-02-12	
2-ST-101-06	3.91	2.29	—	—	3.63	—	—	—	2026-02-12	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2006-02-24	5.13	5.03	0.020	0.010	>50	2026-02-12	
3-CW-101-02	5.49	2.79	2006-02-24	5.20	5.01	0.020	0.019	>50	2026-02-12	
3-CW-101-03	5.49	2.79	2006-02-24	5.08	4.93	0.030	0.015	>50	2026-02-12	
3-CW-101-04	5.49	2.79	2006-02-24	5.38	5.26	0.010	0.012	>50	2026-02-12	
3-CW-101-05	5.49	2.79	2006-02-24	5.33	5.35	0.007	-0.002	>50	2026-02-12	
3-CW-101-06	5.49	2.79	—	—	5.08	—	—	—	2026-02-12	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2006-02-24	5.20	5.06	0.022	0.014	>50	2026-02-12	
3-CW-102-02	5.49	2.79	2006-02-24	5.02	4.37	0.057	0.065	24.2	2026-02-12	
3-CW-102-03	5.49	2.79	2006-02-24	5.07	4.54	0.050	0.053	32.9	2026-02-12	
3-CW-102-04	5.49	2.79	2006-02-24	5.03	4.69	0.041	0.034	46.2	2026-02-12	
3-CW-102-05	5.49	2.79	—	—	4.66	—	—	—	2026-02-12	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2016-02-12	_____